

1. Problem statement & requirements

In this project, I am simulating a basic E-commerce platform. In this platform there are several things that need to be defined first:

- Seller: A person who sells different products on the platform
- Product: An object that is sold by a seller and bought by a buyer
- Buyer: A person who buys products sold on the platform

Requirements:

- Product Catalog: Easy sorting, categories.
- Shopping Cart & Viewing: Viewing products or add product into shopping cart.
- Payment: Can add multiple card but each corresponding to 1 address and 1 phone number.
- Order History: Every completed checkout is recorded permanently, including which card was charged and the price paid at the time of purchase.

Use Case:

- Seller wants to sell products
- Buyer wants to buy products of some category
- A product may be carried by more than one seller, each with their own stock level (marketplace model)
- A buyer's cart, once checked out, becomes a permanent order record

2. Identify the Entities and Relationships

Strong entities

Buyer

Column	Keys
BID	PK
Name	

Seller

Column	Keys
SID	PK
Store_name	
Email	
Phone	

Product

Column	Keys
PID	PK
Product_name	
Category	
Price	

Order

Column	Keys
OrderID	PK
BID	FK
CardID	FK
Order_date	
Total	

Weak entities

Payment_Card

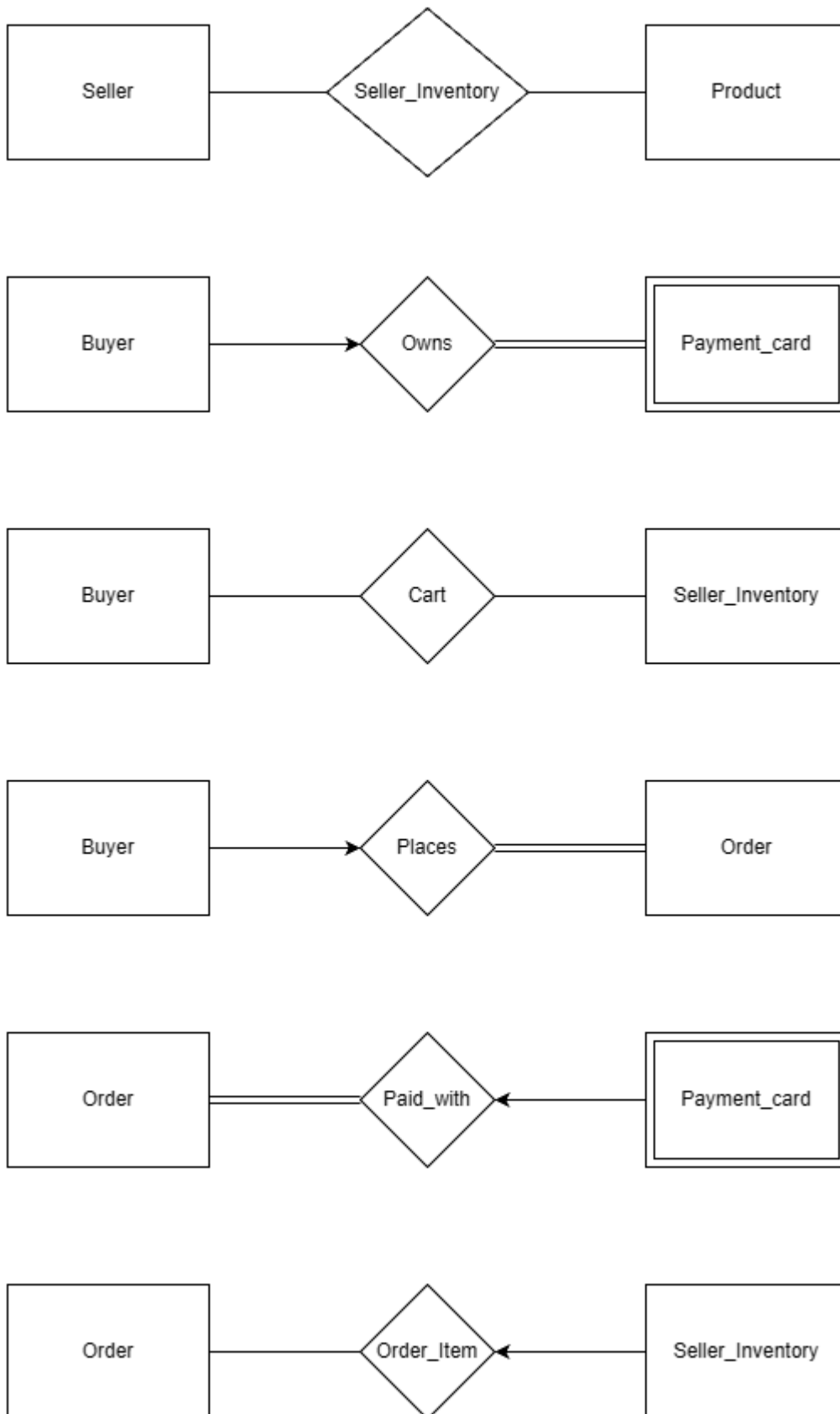
Column	Keys
CardID	PK (partial key)
BID	FK
Card_Number	
Address	
Phone	

Relationships

#	Relationship name	Cardinality	Participants	Attributes	Notes
1	Seller_Inventory	M:N	Seller ↔ Product	(SID FK, PID FK, Quantity, PRIMARY KEY(SID, PID))	Relationship-with-attribute (Quantity). A seller can stock many products; a product

#	Relationship name	Cardinality	Participants	Attributes	Notes
					can be stocked by many sellers.
2	Owns	1:N	Buyer → Payment_Card		A Buyer can have many Payment_Card but each Payment_Card can have 1 Buyer .
3	Cart	M:N	Buyer ↔ Seller_Inventory	(BID FK, SID FK, PID FK, Quantity, PRIMARY KEY(BID, SID, PID), FOREIGN KEY(SID, PID) REFERENCES Seller_Inventory(SID, PID))	A Seller can sell to many different Buyer , vice versa. Each line references one buyer and one specific seller's listing.
4	Places	1:N	Buyer → Order		A Buyer can place many Order ; each order belongs to exactly one buyer .
5	Paid_with	N:1	Order → Payment_Card		Each Order is charged to exactly one of the buyer's Payment_Card but a Buyer's Payment_Card can buy many Order .
6	Order_Item	M:N	Order ↔ Seller_Inventory	(OrderID FK, SID FK, PID FK, Quantity, Price_at_purchase, PRIMARY KEY(OrderID, SID, PID), FOREIGN KEY(SID, PID) REFERENCES Seller_Inventory(SID, PID))	Buyer can have many Order from many Seller_Inventory and vice versa. Permanent record of what Cart looked like at checkout.

Entity-Relationship summary



Relational Schemas

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Buyer(BID PK, Name)
Seller(SID PK, Store_name, Email, Phone)
Product(PID PK, Product_name, Category, Price)
Seller_Inventory(SID FK, PID FK, Quantity, PRIMARY KEY(SID, PID))
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Payment_Card(CardID PK, BID FK, Card_Number, Address, Phone)
Cart(BID FK, SID FK, PID FK, Quantity, PRIMARY KEY(BID, SID, PID),
    FOREIGN KEY(SID, PID) REFERENCES Seller_Inventory(SID, PID))
Order(OrderID PK, BID FK, CardID FK, Order_date, Total)
Order_Item(OrderID FK, SID FK, PID FK, Quantity, Price_at_purchase,
    PRIMARY KEY(OrderID, SID, PID),
    FOREIGN KEY(SID, PID) REFERENCES Seller_Inventory(SID, PID))
```